Nutrition

20 questions: 7 multiple choice, 13 structured.

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1 · Part 2

Paper 1 & 2 Nutrition [1]

Which component of the diet is needed to make haemoglobin in red blood cells?

A calcium

B iron

C vitamin D

D dietary fibre

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2 · Part 2

Paper 1 & 2 Nutrition [1]

Most of the chloroplasts in a leaf are in the palisade mesophyll layer, just below the upper epidermis.

Why is this position an advantage for photosynthesis?

A The palisade layer is the furthest layer from the light, so the chloroplasts are protected from damage.

B The palisade layer is the warmest part of the leaf, so the enzymes work fastest.

C The palisade cells have no cell walls, so light reaches the chloroplasts more easily.

D The palisade cells are nearest the upper surface, so they receive the most light.

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3 · Part 2

Paper 1 & 2 Nutrition [1]

Starch is digested to glucose in two stages, each catalysed by a different enzyme.

Which enzymes are used, in the correct order?

A amylase, then maltase

B maltase, then amylase

C amylase, then lipase

D protease, then maltase

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4 · Part 2

Paper 1 & 2 Nutrition [1]

Bile is alkaline.

Why is this important for digestion in the small intestine?

A It breaks lipids down chemically into smaller molecules.

B It kills all of the bacteria that enter the small intestine.

C It neutralises acid from the stomach, so the pH is suitable for the enzymes in the small intestine.

D It makes the contents of the small intestine more acidic, so that protein digestion can continue.

5 · Part 2

Paper 1 & 2 Nutrition [1]

Which statement describes how food is moved through the gut by peristalsis?

A Muscles along the whole length of the gut contract at the same time.

B Waves of muscle contraction in the gut wall squeeze the food along.

C Gravity pulls the food down the gut, and muscles contract only in the stomach.

D Villi in the small intestine beat in waves to push the food along.

6 · Part 2

Paper 1 & 2 Nutrition [1]

The daily energy requirement of a man is 10 500 kJ. He starts work as a builder, and his daily energy requirement rises to 13 650 kJ.

percentage change = (change in value ÷ original value) × 100

What is the percentage increase in his daily energy requirement?

A 30%

B 23%

C 130%

D 0.30%

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7 · Part 2

Paper 2 only

Nutrition

[1]

Students compare the energy content of different foods. They burn a sample of each food under a boiling tube of water and measure the temperature rise of the water.

Which variable must be kept the same to make this a fair comparison?

A the type of food

B the temperature rise of the water

C the volume of water in the boiling tube

D the energy content of the food

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8 · Part 2

Paper 1 & 2 Nutrition [8]

Green plants make their own food by photosynthesis.

(a) State the word equation for photosynthesis. [2]

(b) Give the balanced chemical symbol equation for photosynthesis. [2]

(c) Name the green pigment in chloroplasts that absorbs light energy for photosynthesis. [1]

(d) Describe the energy conversion that takes place in photosynthesis, and explain why this conversion is important for the plant. [3]

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9 · Part 2

Paper 1 & 2 Nutrition [8]

Scientists grew tomato plants in two growth chambers kept at 25 °C. The air in one chamber contained 0.04% carbon dioxide. The air in the other chamber contained 0.10% carbon dioxide. They measured the rate of photosynthesis at different light intensities.

The table shows their results.

Light intensity / arbitrary units	Rate of photosynthesis at 0.04% carbon dioxide / arbitrary units	Rate of photosynthesis at 0.10% carbon dioxide / arbitrary units
0	0	0
10	12	12
20	22	24
30	28	36
40	30	45
50	30	48

(a) Describe the effect of light intensity on the rate of photosynthesis at 0.04% carbon dioxide. Use data from the table in your answer. [2]

(b) Explain why increasing the light intensity from 40 to 50 arbitrary units does not change the rate of photosynthesis at 0.04% carbon dioxide. [2]

(c) At a light intensity of 10 arbitrary units, the rate of photosynthesis is the same in both chambers. Explain why. [2]

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(d) Calculate the percentage increase in the rate of photosynthesis at a light intensity of 50 arbitrary units when the carbon dioxide concentration is increased from 0.04% to 0.10%.

Use the formula: percentage change = (change in value ÷ original value) × 100 [2]

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10 · Part 2

Paper 1 & 2 Nutrition [7]

In an investigation, a student placed equal volumes of a suspension of a single-celled alga in water baths at different temperatures. The alga carries out photosynthesis. The light intensity was the same for every sample. The student measured the volume of gas released by each sample in 1 hour.

Temperature / °C	Volume of gas released in 1 hour / mm^3
10	8
20	19
30	34
40	15
50	2

(a) Describe the effect of temperature on the volume of gas released. Use data from the table in your answer. [2]

(b) Explain why the volume of gas released increases between 10 °C and 30 °C. [2]

(c) Explain why the volume of gas released decreases between 30 °C and 50 °C. [2]

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(d) Suggest how the student could find the optimum temperature for photosynthesis in this alga more precisely. [1]

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11 · Part 2

Paper 1 & 2 Nutrition [6]

Leaves are adapted for photosynthesis.

(a) Explain how each of these features of a leaf helps photosynthesis.

(i) The leaf is broad and flat. [1]

(ii) The leaf is thin. [1]

(iii) The cells of the upper epidermis are transparent. [1]

(iv) The leaf has a network of veins that contain xylem. [1]

(b) Plants of the same species were grown in two places. One place was in full sunlight. The other place was in deep shade under trees. The plants growing in shade had larger leaves than the plants growing in full sunlight.

Suggest why having larger leaves is an advantage to plants growing in shade. [2]

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12 · Part 2

Paper 1 & 2 Nutrition [8]

Some seedlings of the same species and similar size were grown for four weeks in three different culture solutions. Each solution contained water and mineral ions. The seedlings were then dried and weighed.

Culture solution	Mean dry mass per plant / g	Appearance of leaves
complete (all mineral ions)	12.5	dark green
lacking nitrate ions	4.0	pale green, small
lacking magnesium ions	7.5	yellow

(a) Calculate the percentage decrease in mean dry mass of the plants grown without nitrate ions, compared with the plants grown in the complete solution.

Use the formula: percentage change = (change in value ÷ original value) × 100 [2]

(b) Explain why the plants grown without nitrate ions had the lowest mean dry mass. [3]

(c) Explain why the plants grown without magnesium ions had a lower mean dry mass than the plants grown in the complete solution. [3]

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13 · Part 2

Paper 1 & 2 | Nutrition | [7]

Jamal investigated the effect of light intensity on the rate of photosynthesis in a water plant.

He placed a piece of pondweed in a boiling tube containing sodium hydrogencarbonate solution. He placed a lamp 10 cm from the boiling tube and waited 5 minutes. He then counted the number of bubbles of gas released from the cut end of the stem in 1 minute. He repeated this with the lamp at 20 cm, 30 cm and 40 cm from the boiling tube.

(a) State the independent variable and the dependent variable in Jamal's investigation. [2]

(b) Sodium hydrogencarbonate releases carbon dioxide into the water. Explain why Jamal used sodium hydrogencarbonate solution. [2]

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(c) The lamp also gives out heat. Explain how this could affect Jamal's results, and suggest how he could prevent this. [2]

(d) Counting bubbles is not an accurate way to measure the rate of photosynthesis. Suggest a more accurate way to measure the gas released. [1]

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14 · Part 2

Paper 1 & 2 | Nutrition | [9]

Leila kept a potted plant with green leaves in the dark for 48 hours. She then covered part of one leaf, on both sides, with a strip of black paper that does not let light through. She placed the plant in bright light for 24 hours. She then removed the leaf from the plant, took off the black paper and tested the whole leaf for starch.

(a) Explain why Leila kept the plant in the dark for 48 hours before the experiment. [2]

(b) Describe how Leila should test the leaf for starch. Include any safety precautions she should take. [4]

(c) Predict the colour of the part of the leaf that was covered and the colour of the part that was uncovered after the test. Explain your answer. [3]

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15 · Part 2

Paper 1 & 2 Nutrition [7]

The table shows the nutrients in 100 g of three foods.

Food	Protein / g	Lipid / g	Carbohydrate / g	Dietary fibre / g	Calcium / mg	Vitamin C / mg
brown rice (cooked)	2.6	0.9	23.0	1.8	10	0
cheese	25.0	33.0	1.3	0	720	0
oranges	0.9	0.1	11.8	2.4	40	53

(a) Identify the food in the table that would be best for building strong bones and teeth. Give a reason for your answer. [2]

(b) State one function of vitamin C in the body. [1]

(c) Calculate the mass of protein in a 60 g portion of cheese. [2]

(d) Explain why a person who ate a diet containing mainly cheese could develop constipation. [2]

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16 · Part 2

Paper 2 only

Nutrition

[8]

The table shows the daily energy requirements of four people. All four people have similar, moderate activity levels.

Person	Daily energy requirement / kJ
girl aged 8	7 400
girl aged 15	9 000
woman aged 30	8 400
woman aged 30, in the last three months of pregnancy	9 200

(a) Explain why the girl aged 15 needs more energy each day than the woman aged 30. [2]

(b) Explain why the pregnant woman needs more energy each day than the woman who is not pregnant. [2]

(c) A student investigated the energy content of a breakfast cereal. She burned a 0.80 g sample of the cereal under a boiling tube containing 25 cm^3 of water. The temperature of the water rose by 12 °C.

Use the formula: energy transferred to the water (J) = mass of water (g) × 4.2 × temperature rise (°C)

1 cm^3 of water has a mass of 1 g.

Calculate the energy released per gram of cereal in this investigation. Give your answer in J per g. [2]

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(d) The packet states that 1 g of this cereal provides 16 kJ of energy.

Calculate the mass of cereal that would provide the extra energy needed each day by the pregnant woman, compared with the woman aged 30 who is not pregnant. Use the value on the packet. [2]

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17 · Part 2

Paper 1 & 2 Nutrition [7]

The human alimentary canal is a long tube that runs from the mouth to the anus.

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(a) Name the part of the alimentary canal where each of these happens.

(i) Water is absorbed from the undigested food, so that the faeces become more solid. [1]

(ii) Faeces are stored before egestion. [1]

(iii) Food passes from the mouth to the stomach. [1]

(b) Bile is needed for digestion. State the organ that produces bile and the organ that stores bile. [2]

(c) The pancreas is connected to the alimentary canal, but food does not pass through it. Describe the role of the pancreas in digestion. [2]

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18 · Part 2

Paper 1 & 2 Nutrition [8]

Sam investigated the digestion of lipid in milk. He set up three tubes. Each tube contained 5 cm^3 of milk and 7 cm^3 of sodium carbonate solution, so that every tube started at pH 9.0. He then added the substances shown in the table. All the tubes were kept at 35 °C. He recorded the time taken for the pH in each tube to fall to 7.0.

Tube	Substances added	Time taken for pH to fall to 7.0 / minutes
1	1 cm^3 lipase solution and 1 cm^3 bile salts solution	4
2	1 cm^3 lipase solution and 1 cm^3 water	11
3	1 cm^3 boiled and cooled lipase solution and 1 cm^3 bile salts solution	pH did not fall in 30 minutes

(a) Explain why the pH fell in tube 1. [2]

(b) Explain why the pH fell faster in tube 1 than in tube 2. [3]

(c) Explain the result for tube 3. [2]

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(d) Give one reason why Sam kept all the tubes at the same temperature. [1]

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19 · Part 2

Paper 1 & 2 Nutrition [8]

The inner wall of the small intestine is covered with millions of tiny finger-like projections called villi. Each villus has a wall that is one cell thick, a dense network of blood capillaries, and a central vessel that absorbs the products of lipid digestion. The epithelial cells on the surface of each villus have microvilli.

(a) State the name of the central vessel in a villus that absorbs the products of lipid digestion. [1]

(b) Explain how the small intestine and its villi are adapted for the efficient absorption of digested food. [5]

(c) In coeliac disease, the villi become damaged and flattened. Suggest why a person with untreated coeliac disease may lose body mass, even when eating a balanced diet. [2]

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20 · Part 2

Paper 2 only

Nutrition

[8]

Amira investigated the energy content of peanuts.

She put 20 cm^3 of water into a boiling tube held in a clamp and recorded the temperature of the water. She pushed a peanut of mass 0.50 g onto a mounted needle and set it alight using a Bunsen burner. She then held the burning peanut under the boiling tube until it stopped burning, and recorded the highest temperature of the water. She repeated this with two more peanuts, each of mass 0.50 g.

Trial	Temperature rise of water / °C
1	27
2	29
3	31

(a) Calculate the mean temperature rise of the water. [1]

(b) Using her mean value, Amira calculated that the energy content of the peanuts was 4.9 kJ per g. The packet states that the energy content of the peanuts is 25 kJ per g.

Calculate Amira's value as a percentage of the value on the packet.

Use the formula: percentage = (Amira's value ÷ value on the packet) × 100 [2]

(c) Explain why Amira's value is much lower than the value on the packet. [3]

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(d) Suggest two improvements to Amira's method that would give a value closer to the value on the packet. [2]

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